RUNTIME TYPOLOGY: RIGID VS ELASTIC BASELINE

Paper III

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Research Series: Voigtian Field Research Series • Volume III
Type: Scientific Paper / Runtime Typology
Version: v1.0 Canonical Research Draft
Recommended Distribution: Zenodo • SSRN • ResearchGate

"The Voigtian Field transforms invisible degradation into empirical laws for living baseline reconstruction."

ABSTRACT

This paper establishes a formal typology of computational environments based on **drift reconstruction speed, baseline depth, and elastic recovery signature after controlled perturbations**.

The central proposition is that distinct runtime environments preserve and reconstruct baseline coherence at different rhythms, forming **reproducible runtime classes**.

This classification introduces a new lens for workload governance, intelligent cloud placement, and predictive operational stability.

INTRODUCTION

Different environments preserve and reconstruct baseline coherence at different temporal rates.

The central hypothesis is that this difference forms reproducible runtime classes.

Rather than treating all systems as homogeneous operational entities, this work proposes that baseline behavior itself defines a structural taxonomy.

METHODOLOGY

The following environments were compared:

- headless Linux server in Munich
- Windows desktop with intense visual concurrency
- rest cycles, progressive stress, and recovery phases
- multi-day observed temporal series

CLASS A — RIGID BASELINE RUNTIME

Characteristics

- slow drift return
- high inertia
- greater baseline depth
- prolonged persistence at **0.92** before returning to **1**
- excellent longitudinal stability

Implications

Ideal for:

- critical APIs
- stable clusters
- financial services
- persistent edge nodes

CLASS B — ELASTIC BASELINE RUNTIME

Characteristics

- fast drift return
- high plasticity
- rapid working set recovery
- temporary elastic reserve
- higher sensitivity to successive bursts

Implications

Ideal for:

- workstations
- development environments
- interactive workloads
- desktops with high context switching

CONCLUSION

The **rigid vs elastic baseline typology** offers a new framework for workload classification, intelligent cloud placement, and runtime-aware infrastructure design.

This paper expands the Voigtian Field from empirical laws into **runtime taxonomy science**.

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Dados: 2026.04.09 14:31:22 -03'00'